

Tutorial 6 – Securing IS: The Case of SingHealth

Objective

In this tutorial, we examine Singapore's most serious cybersecurity breach — the SingHealth attack — to understand how system vulnerabilities and poor security practices can lead to large-scale data loss. Students will analyse the incident using key IS security concepts introduced in class.

Duration

1 – 1.5 hours

Activity Instructions

1. Read Article (15 minutes)

Students should read the article:

[Singapore's privacy watchdog fines IHiS, SingHealth for data breach.](#)

This article provides a real-world account of Singapore's most serious cybersecurity breach. It highlights the key events, organizational failings, and the regulatory response following the SingHealth incident. While reading, students should focus on:

- How the breach happened
- What vulnerabilities were exploited
- What fines and actions were taken by the authorities
- The lessons learned for IS infrastructure and security governance

2. Group Discussion (30 minutes)

In your project group, discuss the following questions:

a. What types of information security threats were present in the SingHealth cyberattack?

Hint: Refer to categories like software vulnerabilities, unauthorized access, lack of intrusion detection, and possible social engineering.

b. Why was SingHealth's information system vulnerable to attack? (*L8 pp.10-11*)

c. Apply the *CIA triad* (*L8 pp.16-17*) to evaluate the impact of this breach:

- How was Confidentiality compromised?
- Was Integrity affected?
- Was Availability ever threatened?

d. What technical safeguards (*L8 pp.31*) could have helped prevent or limit this breach?

- Encryption
- Multi-factor Authentication

- Intrusion Detection

3. Individual Submission (Post-Class)

1. Each student must individually submit answers to the questions in a **PDF file of no more than two pages**, named:
T6_<full-name>.pdf(e.g., T6_xiaoming.pdf).
2. Students must submit their work **before the next tutorial** to receive the **full score**.
3. **Late submissions will receive only half the score. Submissions later than 2 weeks will not receive any score.**

Quick Reading Aid:

Background

In July 2018, Singapore faced its most serious cyberattack when the personal data of 1.5 million patients — including Prime Minister Lee Hsien Loong — was stolen from the healthcare group SingHealth. The attack exploited vulnerabilities in the system managed by the Integrated Health Information Systems (IHIS), exposing major lapses in cybersecurity practices, monitoring, and response. The incident triggered public outcry and a full investigation by the Personal Data Protection Commission (PDPC), Singapore's data privacy regulator.